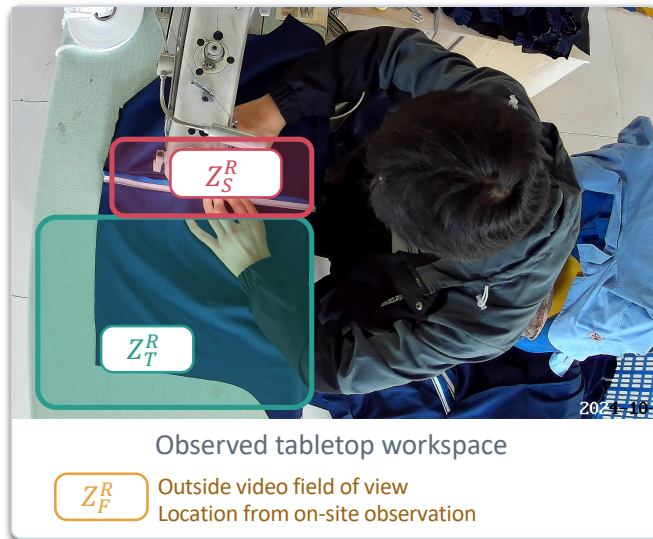


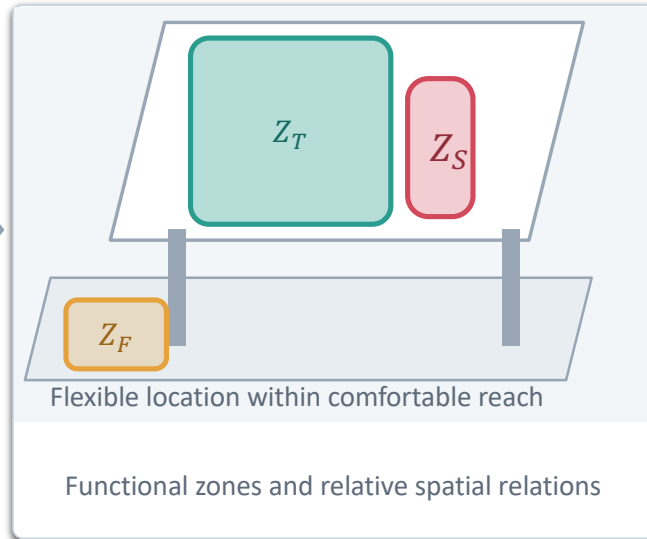
Spatial translation from the real sewing workstation to the VR-WCE

Functional relations were preserved through geometric abstraction rather than exact metric replication.

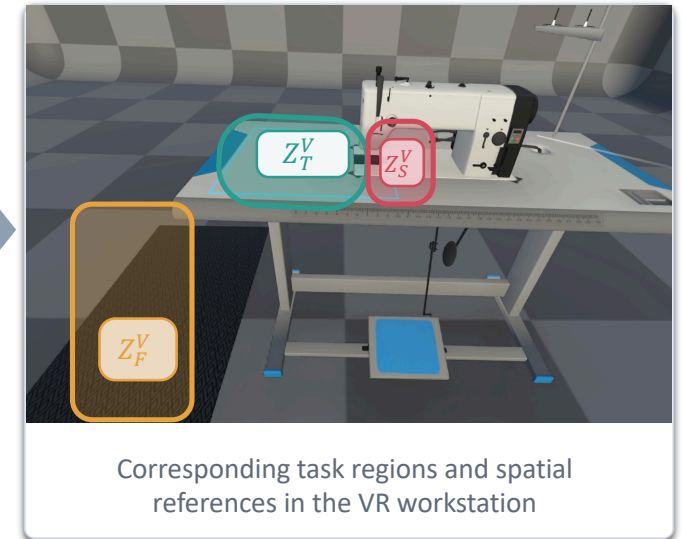
A Real workplace



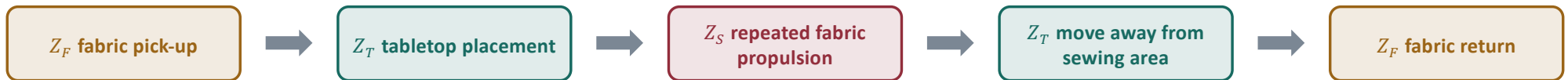
B Functional spatial abstraction



C VR implementation



Preserved material-flow sequence



Z_F : floor-level material-exchange zone Z_T : tabletop handling zone Z_S : needle-adjacent sewing zone